CSI131 Discrete Structures I

Tutorial Mathematical Induction

Answer ALL questions

1. Prove by mathematical induction that
$1 + 3 + 5 + \cdots + (2n - 1) = n^2$, $\forall n \geq 1$.
Use the template discussed in the class.

2. Prove by mathematical induction that
$1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $\forall n \geq 1$.
Use the template discussed in the class.

3. Prove by mathematical induction that
$1.3 + 2.4 + 3.5 + 4.6 + \cdots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$, $\forall n \geq 1$.
Use the template discussed in the class.

5. Prove by mathematical induction that
$1^3 + 2^3 + 3^3 + 4^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$, $\forall n \geq 1$.
Use the template discussed in the class.

6. Prove by mathematical induction that
$1 + 2 + 4 + 8 + \cdots + (2^{n-1}) = 2^n - 1$, $\forall n \geq 1$.
Use the template discussed in the class.

7. Prove by mathematical induction that
$2^1 + 2^2 + 2^3 + \cdots + 2^n = 2^{n+1} - 2$, $\forall n \geq 1$.
Use the template discussed in the class.

8. Prove by mathematical induction that
$1.1! + 2.2! + 3.3! + \cdots + n.n! = (n+1)! - 1$, $\forall n \geq 1$.
Use the template discussed in the class.

9. Prove by mathematical induction that
$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$, $\forall n \geq 1$.
Use the template discussed in the class.